



Sort and filter data to keep it organized

The first two phases of data analysis, **Organize data and Format and adjust data**, are important for data analysts because **they can use these phases to manipulate their data in ways that make important patterns and trends more obvious**. Most of the datasets you'll use as a data analyst will be organized as tables. Tables are helpful because they let you **manipulate** and categorize your data. Having **distinct** categories and classifications lets you focus on, and differentiate between, the groups in your data quickly and easily.

Sorting and filtering are two methods you can use to organize, format, and adjust data. For example, **a filter can help you find errors or outliers so you can fix or flag them before your analysis**. **Outliers are data points that are very different from similarly collected data and might not be reliable values**. **The benefit of filtering the data is that after you fix errors or identify outliers, you can remove the filter and return the data to its original organization**.

In this reading, you'll review sorting and filtering and consider how they can be used together. You'll also be introduced to how a particular form of sorting is done in a pivot table.

Sort data

Sorting is the process of arranging data into a meaningful order to make it easier to understand, analyze, and visualize. It ranks your data based on a specific metric you

choose. You can sort data in spreadsheets, SQL databases (when your dataset is too large for spreadsheets), and tables in documents.

To rank items or create chronological lists, you can sort by ascending or descending order. **Sorting arranges the data in a meaningful way and gives you immediate insights. Sorting also helps you to group similar data together by a classification.** For example, if a ski resort design company wants to evaluate the resorts designed by a competitor, a data analyst can sort competitive resorts by locations, runs, **acreage**, and other factors. This way, the firm's designers can visit the types of resorts they also design and gather information that could be used in its own future designs.

The screenshot shows a Google Sheet titled "Ski Resorts Portfolio Project". The "Data" menu is open, displaying options like "Sort sheet", "Sort range", "Create a filter", "Filter views", "Add a slicer", "Protect sheets and ranges", "Named ranges", "Named functions", "Randomize range", "Column stats", "Data validation", "Data cleanup", "Split text to columns", "Data extraction", and "Data connectors". The "Sort sheet" option is selected, showing a sub-menu with "Sort sheet by column A (A to Z)" and "Sort sheet by column A (Z to A)". The spreadsheet data is as follows:

	A	B		F	G	H
	resort_name	state/terri		lifts	runs	acres
1	resort_name	state/terri				
2	Alyeska Resort	Alaska		41	7	76
3	Eaglecrest Ski Area	Alaska		72	4	37
4	Hilltop Ski Area	Alaska		11	3	13
5	Castle Mountain	Alberta		63	6	94
6	Lake Louise	Alberta		91	11	164
7	Marmot Basin	Alberta		14	7	91
8	Mt. Norquay	Alberta		05	6	60
9	Nakiska Ski Area	Alberta		35	6	75
10	Sunshine Village	Alberta		70	11	145
11	WinSport	Alberta		20	7	6
12	Caviahue	Argentina		18	13	22
13	Cerro Catedral Alta Patagonia	Argentina		50	29	55
14	Chapelco	Argentina		67	12	28
15	Las Lenas	Argentina		90	15	30
16	Arizona Snowbowl	Arizona		04	8	55
17	Sunrise Park Resort	Arizona		12	8	65
18	Charlotte Pass	Australia		99	5	23
19						
20						
21						
22						
23						
24						

An example of sorting a spreadsheet of ski resorts, including information about resort name, state/territory/country, lifts, runs, and acres. The image taker has clicked into the

Data menu option, selected Sort sheet, and is hovering over Sort sheet by column A (A to Z).

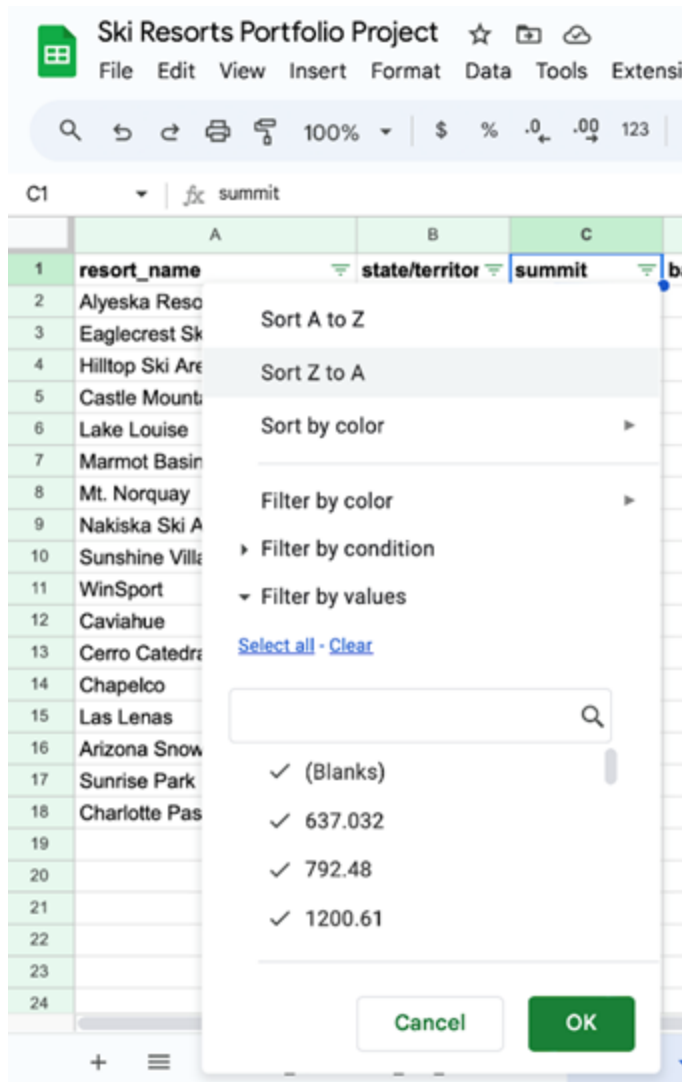
Filter data

Sometimes, an analysis may require only a subset of the data in your dataset. You can use a filter to show only the data that meets a specified criteria while hiding the rest. Filtering is useful when you have lots of data. You can save time by zeroing in on the data that's important for your current analysis or the data that contains errors. Most spreadsheets and SQL databases allow you to filter your data in a variety of ways. Filtering gives you the ability to find what you are looking for without too much effort.

For example, if the ski resort design company wants to inspect specific criteria for the competitive ski resorts they intend to visit and evaluate, a data analyst can filter the competitive resort database to extract information about the number of runs compared to acreage to identify design trends or other insights.

The screenshot shows a Google Sheets interface for a project titled "Ski Resorts Portfolio Project". The "Data" menu is open, and the "Create a filter" option is highlighted. The spreadsheet contains data for 18 ski resorts, with columns for resort_name, state/territory, lifts, runs, acres, green_percent, blue_percent, and black_percent. The data is as follows:

	A	B	F	G	H	I	J	K
	resort_name	state/territory	lifts	runs	acres	green_percent	blue_percent	black_percent
1	Alyeska Resort	Alaska	41	7	76	1610	11	52
2	Eaglecrest Ski Area	Alaska	72	4	37	640	32	38
3	Hilltop Ski Area	Alaska	11	3	13	30	80	10
4	Castle Mountain	Alberta	63	6	94	3592	10	21
5	Lake Louise	Alberta	91	11	164	4200	25	45
6	Marmot Basin	Alberta	14	7	91	1720	30	30
7	Mt. Norquay	Alberta	05	6	60	190	38	25
8	Nakiska Ski Area	Alberta	35	6	75	1021	15	59
9	Sunshine Village	Alberta	70	11	145	3358	18	26
10	WinSport	Alberta	20	7	6	214	33	33
11	Caviahue	Argentina	18	13	22	1494	30	40
12	Cerro Catedral Alta Patagonia	Argentina	50	29	55	1480	4	51
13	Chapelco	Argentina	67	12	28	140	25	25
14	Las Lenas	Argentina	90	15	30	43000	15	45
15	Arizona Snowbowl	Arizona	04	8	55	777	22	44
16	Sunrise Park Resort	Arizona	12	8	65	1200	46	28
17	Charlotte Pass	Australia	99	5	23	124	32	32

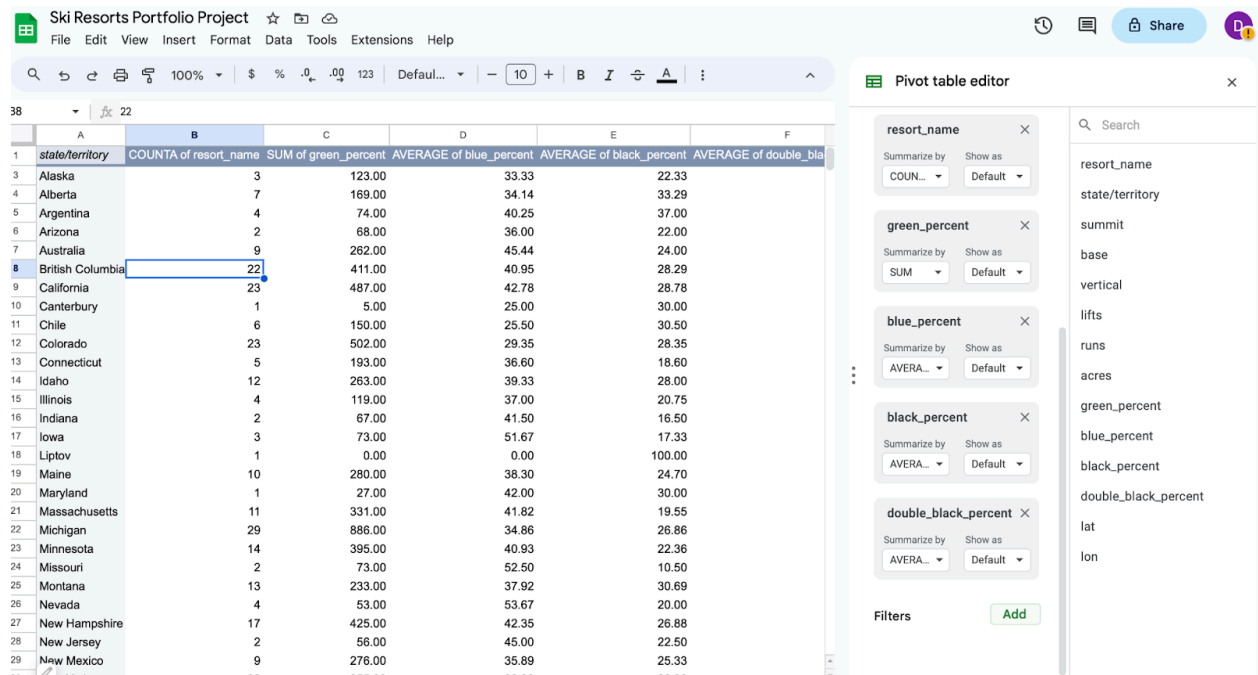


An example of filtering data in a spreadsheet of ski resort information by specific evaluation criteria such as location, acreage, or number of runs

Sort a pivot table

A pivot table is a data summarization tool used to sort, reorganize, group, count, total, or average data. Items in the row and column areas of a pivot table are sorted in ascending order by any custom list first. If the items aren't in a custom list, they will be sorted in ascending order by default. But, if you sort in descending order, you are setting up a rule that controls how the field is sorted even after new data points are added. For

example, in the ski resort dataset, the pivot table allows locations to be sorted alphabetically by state, territory, or country.



The screenshot shows a Google Sheets interface with a pivot table titled "Ski Resorts Portfolio Project". The pivot table is sorted by state/territory. The pivot table editor is open on the right, showing the configuration for the pivot table.

state/territory	COUNTA of resort_name	SUM of green_percent	AVERAGE of blue_percent	AVERAGE of black_percent	AVERAGE of double_black_percent
Alaska	3	123.00	33.33	22.33	
Alberta	7	169.00	34.14	33.29	
Argentina	4	74.00	40.25	37.00	
Arizona	2	68.00	36.00	22.00	
Australia	9	262.00	45.44	24.00	
British Columbia	22	411.00	40.95	28.29	
California	23	487.00	42.78	28.78	
Canterbury	1	5.00	25.00	30.00	
Chile	6	150.00	25.50	30.50	
Colorado	23	502.00	29.35	28.35	
Connecticut	5	193.00	36.60	18.60	
Idaho	12	263.00	39.33	28.00	
Illinois	4	119.00	37.00	20.75	
Indiana	2	67.00	41.50	16.50	
Iowa	3	73.00	51.67	17.33	
Liptov	1	0.00	0.00	100.00	
Maine	10	280.00	38.30	24.70	
Maryland	1	27.00	42.00	30.00	
Massachusetts	11	331.00	41.82	19.55	
Michigan	29	886.00	34.86	26.86	
Minnesota	14	395.00	40.93	22.36	
Missouri	2	73.00	52.50	10.50	
Montana	13	233.00	37.92	30.69	
Nevada	4	53.00	53.67	20.00	
New Hampshire	17	425.00	42.35	26.88	
New Jersey	2	56.00	45.00	22.50	
New Mexico	9	276.00	35.89	25.33	

Image of a pivot table of the ski resort data, with the pivot table editor open and ready for parameters to be entered. The data is grouped by state, territory, or country.

Key takeaways

Data analysts filter and sort data to organize it for better understanding, analysis, and visualization. Sorting arranges data in a meaningful order, while filtering displays only data that meets specific criteria. Combining filtering and sorting allows for organizing only relevant data for analysis. Both spreadsheets and SQL databases allow for data filtering and sorting data.